Detecting Concept Drift In Process Mining Using Explainable AI

Submitted in partial fulfillment of the requirements for the award of the degree of

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Information Technology

Submitted by

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The forgoing thesis is hereby approved as a credible study in the area of Information Technology and its related areas carried out and presented in a manner satisfactory to warrant its acceptance as a prerequisite to the degree for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, opinion expressed, or conclusion drawn therein but approve the thesis only for the purpose for which it is submitted.

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I hereby declare that the work presented in this report entitled "Detecting Concept Drift In Process Mining Using Explainable AI", submitted to fulfill the MASTER'S THESIS report of M.Tech - IT at Indian Institute of Information Technology, Allahabad, is an authenticated record of our original work carried out under the guidance of Prof. Ranjana Vyas. Due acknowledgments have been made in the text for all other material used. The project was carried out in full compliance with the requirements and constraints of the prescribed curriculum.

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CERTIFICATE FROM SUPERVISOR

This is to certify that the statement made by the candidate is correct to the best of my knowledge and belief. The project titled "Detecting Concept Drift In Process Mining Using Explainable AI" is a record of candidates' work carried out by him under my guidance and supervision. I do hereby recommend that it should be accepted in the fulfillment of the requirements of the MASTER'S THESIS at IIIT Allahabad.

Signature of the Supervisor: ————————————-

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Abstract

In dynamic business environments, business process behaviors are subject to frequent changes driven by internal and external factors, resulting in concept drift. Traditional process mining techniques often overlook such drifts, leading to outdated or misleading process models. This thesis presents a novel, explainable framework for detecting concept drift in business process event logs by integrating Explainable Artificial Intelligence (XAI) techniques. The proposed approach leverages machine learning models to identify changes across control-flow, organizational, and temporal dimensions of process execution. By embedding explainability, the framework not only detects when and where drift occurs but also provides interpretable insights into the underlying causes, supporting data-driven decision-making and continuous process improvement. Experimental evaluation on both synthetic and real-world event logs demonstrates the effectiveness of the approach in achieving timely, accurate, and transparent drift detection. This work advances adaptive process mining by narrowing the gap between predictive performance and human interpretability, ultimately enabling more resilient and intelligent business process management.

KEYWORDS : Concept Drift, Process Mining, Explainable AI (XAI), Event Logs, Drift Detection, Business Process Analysis, Interpretability

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Abbreviations

PM	Process Mining
PD	Process Discovery
PC	Process Conformance
BPM	Business Process Management
LAP 2017	Loan Application process((BPI) Challenges 2017)
IML 2013	Incident Management Log ((BPI) Challenges 2013)
PMLO 2013	Problem Management Log,open ((BPI) Challenges 2013)
PMLC 2013	Problem Management Log,closed ((BPI) Challenges 2013)
LAP 2012	Loan Application process((BPI) Challenges 2012)
FIT	Fraunhofer Institute for Applied Information Technology

Chapter 1

Introduction

1.1 Background and Motivation

1.1.1 Process Mining: Unlocking Business Process Insights

Process mining is an advanced analytical discipline that bridges the gap between traditional model-based process analysis and data-driven insights. At its core, process mining leverages event logs—digital records of activities generated by information systems—to automatically discover, monitor, and improve real business processes. Unlike conventional business process management, which often relies on static models or subjective interviews, process mining provides objective, data-backed views of how processes actually unfold in practice.

Key capabilities of process mining include:

- **Process Discovery:** Automatically reconstructing process models from event log data, revealing the true flow of activities, decisions, and exceptions.
- **Conformance Checking:** Comparing discovered process models with predefined standards or reference models to identify deviations, compliance issues, and inefficiencies.

- **Performance Analysis:** Measuring process metrics such as throughput times, bottlenecks, and resource utilization directly from operational data.

Process mining has become indispensable for organizations seeking transparency, operational excellence, and continuous improvement. It is widely applied in domains such as healthcare, finance, manufacturing, and customer service, where understanding and optimizing complex processes is critical.

1.1.2 The Challenge of Concept Drift in Dynamic Environments

In today's rapidly changing business landscape, processes are rarely static. They evolve in response to shifting regulations, emerging technologies, changing customer expectations, and internal reorganizations. This evolution often leads to **concept drift**—a phenomenon where the underlying behavior or structure of a business process changes over time.

Concept drift poses significant challenges:

- **Model Obsolescence:** Process models that accurately reflected past behavior may become outdated, leading to poor predictions, suboptimal automation, or compliance risks.
- **Hidden Inefficiencies:** Undetected drifts can introduce inefficiencies, increase costs, or degrade service quality.
- **Process Failures:** In critical domains, such as healthcare or finance, failing to recognize process drift can result in serious errors or regulatory violations.

Traditional process mining techniques typically assume that processes remain stable over time. However, real-world event logs—especially those from dynamic sectors like hospitals, banks, and service centers—often contain evidence of temporal, structural, and performance drifts. These drifts manifest as:

- **Temporal Drift:** Changes in the timing or frequency of activities.
- **Structural Drift:** Alterations in the sequence or flow of activities.
- **Performance Drift:** Shifts in process efficiency, such as increased case duration or resource consumption.

Motivation:
Addressing concept drift is essential for organizations aiming to maintain accurate process models, ensure compliance, and drive continuous improvement. By integrating advanced drift detection and explainability into process mining, organizations can proactively adapt to change, mitigate risks, and sustain operational excellence in an ever-evolving environment.

1.1.3 Types of Concept Drift

Concept drift in business process event logs refers to changes in the underlying behavior or structure of a process over time. These shifts may occur due to evolving business rules, market dynamics, customer behavior, or internal operational changes. Understanding the types of drift is essential for building robust detection mechanisms and ensuring process model relevance.

- **Sudden (Abrupt) Drift:** Characterized by an immediate shift in process behavior, typically following a major event such as a policy change or system update. From the point of change onward, all instances follow the new process variant [4, 5, 6].
- **Gradual Drift:** Occurs when new process behaviors progressively replace the old ones. During the transition, both variants may coexist, making detection more complex [4, 5, 6].

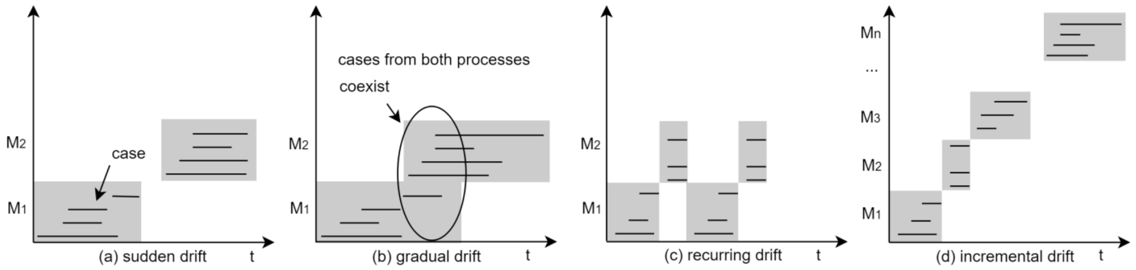

FIGURE 1.1: Types of Concept Drift: (a) Sudden, (b) Gradual, (c) Recurring, (d) Incremental. Adapted from [1].

- **Incremental Drift:** Involves small, successive modifications that accumulate over time. Though each change may be minor, the cumulative effect can significantly alter the process [4, 5].

- **Recurring (Seasonal) Drift:** Refers to patterns that cyclically reappear, such as processes adapting to seasonal demand or periodic policy changes. Unlike sudden or incremental drift, recurring drift returns to previously observed behaviors [4, 5].

- **Performance Drift:** Represents a shift in quantitative process metrics—such as case duration, resource usage, or event density—without structural changes to the process model. This often signals hidden inefficiencies or workload imbalances [4, 6].

- **Structural Drift:** Involves changes in the control-flow of the process, including the appearance or disappearance of paths or activities. Such drift directly alters the process model and may indicate procedural updates or user workarounds [4, 6, 7].

Figure 1.1 provides a visual overview of the most commonly observed types of concept drift in event logs, emphasizing their behavioral differences and transition characteristics.

1.1.4 Types of Concept Drift Considered

This thesis is designed to detect and analyze multiple types of concept drift in business process event logs, specifically focusing on the following:

- **Temporal (Distributional) Drift:** Identified by analyzing changes in activity and resource distributions across time windows using statistical divergence measures such as Jensen-Shannon Divergence.
- **Structural Drift:** Detected by comparing Directly-Follows Graphs (DFGs) across different time segments to identify new, disappeared, or significantly altered process paths.
- **Performance Drift:** Measured by monitoring changes in process performance metrics such as case duration and event complexity over time.

These drift types form the core of this research and provide a comprehensive framework for understanding how business processes evolve. By combining the detection of distributional, structural, and performance drift, this thesis ensures a well-rounded and practical approach to concept drift analysis—addressing both academic rigor and real-world applicability.

1.1.5 Scope Justification

While concept drift can manifest in various forms such as sudden, gradual, incremental, and recurring drift, this work focuses on temporal (distributional), structural, and performance drifts. These are the most impactful and commonly encountered in real-world business process environments. The selected types align with the objectives of process mining and are well-supported by the implemented algorithms and available event log data. Other drift forms, though important, require specialized methods beyond the scope of this thesis and are suggested for future work.

1.2 Problem Statement

Despite recent advances in drift detection algorithms, there is a lack of interpretability and explainability in the results. Most methods quantify drift using statistical or structural metrics (e.g., Jensen-Shannon divergence, Directly-Follows Graphs), but fail to communicate the insights in an accessible and human-understandable manner.

This thesis addresses the dual challenge of:

- Detecting multi-perspective concept drift (activities, resources, performance, structure) in event logs using process mining techniques, and
- Explaining the detected drifts using a large language model (LLM) to enhance interpretability for end users and decision-makers.

1.3 Objectives

The primary objectives of this thesis are:

- To analyze the BPI Challenge 2017 event log and identify patterns of concept drift using PM4Py.
- To detect drift in:
  - Activity and resource distribution (temporal drift)
  - Control-flow variants (DFG drift)
  - Case duration and event complexity (performance drift)
- To generate a comprehensive drift analysis report summarizing findings across time windows.
- To integrate a Gemini LLM API to produce human-readable explanations of the drift reports.

1.4 Contribution and Visual Overview

Contribution

This work presents a semi-automated framework that combines process mining drift detection with Explainable AI (XAI) techniques. While most previous works focus only on quantitative drift metrics, this thesis:

- Produces structured, multi-perspective drift reports.
- Introduces a novel post-hoc explanation layer using LLMs to communicate insights.
- Demonstrates the practical utility of combining process mining with generative AI to support decision-makers, auditors, and process analysts.

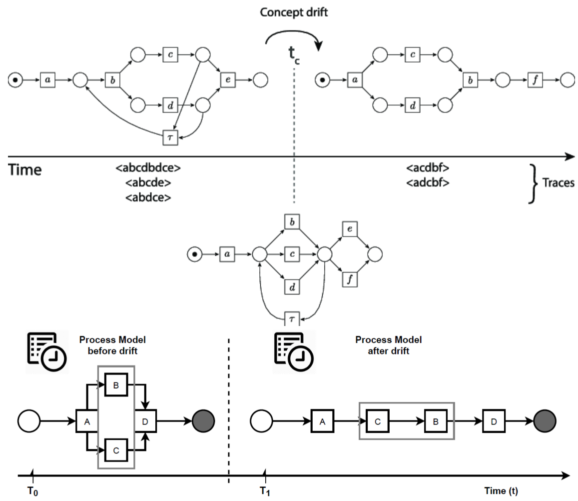

Figure 1: Top – Simple process control-flow drift; Bottom – General concept drift over time.

1.4.1 Explanation of Illustrative Figures

The visualizations included above and below demonstrate various dimensions of concept drift in process mining. These figures help contextualize the multi-perspective analysis that this thesis undertakes.

Figure 1 illustrates two key perspectives:

- The **top diagram** shows a basic workflow before and after process drift. Such drift could occur due to changes in policy, technology, or user behavior. For instance, a patient admission process in a hospital might skip steps under emergency conditions.

- The **bottom diagram** provides a high-level abstraction of how a process's structure evolves over time—indicating the potential for drift between different time windows.

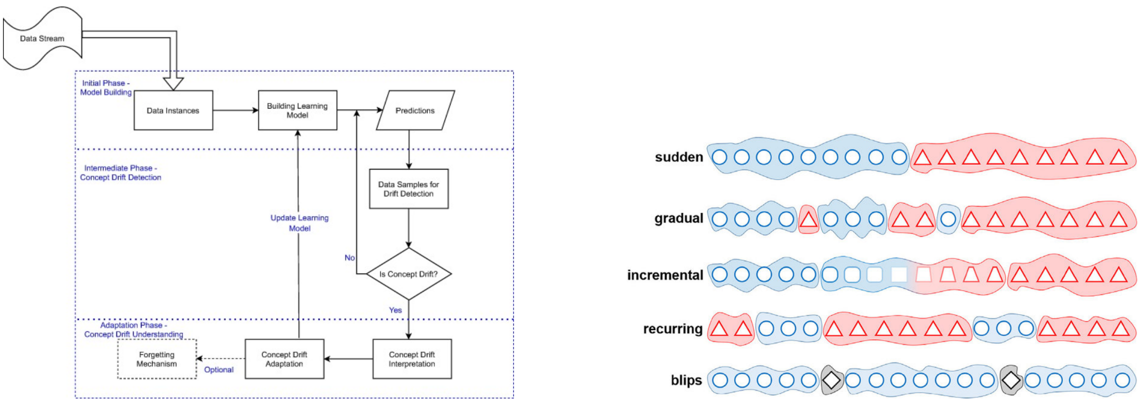

FIGURE 1.2: **Figure 2:** (Left) Process evolution across versions; (Right) Taxonomy of drift types.

Figure 2 further elaborates:

- The **left image** visualizes how control-flow variants emerge across different time windows or case clusters. This is critical when using DFGs for drift detection in event logs.

- The **right image** shows a taxonomy of concept drift: sudden, gradual, recurring, and incremental. Each of these has different detection and handling implications. For example, sudden drifts require fast adaptation, whereas recurring drifts can be anticipated using pattern history.

Together, these figures reinforce the motivation for a framework that not only detects drift but also explains it in a human-understandable way using LLM-based explanations.

Chapter 2

Literature Review

2.1 Concept Drift in Process Mining

Business processes are dynamic, evolving in response to internal and external pressures such as regulatory changes, market shifts, and resource reallocation. In process mining, concept drift refers to any significant change in the underlying process behavior over time, which can manifest as sudden, gradual, recurring, or incremental changes. process mining techniques often assume process stability, making them ill-suited for environments where such drift is common [4] [1].

A comprehensive survey by Sato et al. (2021) highlights that most process mining tools and algorithms were originally designed for offline, static analysis, and only recently have researchers begun to address the challenges of drift detection in dynamic, real-world settings. The review also notes the lack of standardized protocols and benchmark datasets for evaluating drift detection methods, complicating comparative assessment and industrial adoption [4] [1].

2.2 Drift Detection Techniques

Early approaches to drift detection in process mining focused on segmenting event logs into temporal windows and comparing process models or activity distributions between these windows. Techniques such as change point detection, statistical divergence (e.g., Jensen-Shannon divergence), and directly-follows graph (DFG) comparison have been widely adopted [6] [8]. While these methods can identify when and where changes occur, they often provide limited insight into the nature or causes of the drift.

Recent advances have expanded drift detection beyond the control-flow perspective to include resource, performance, and data perspectives. For example, Adams et al. (2023) introduced a framework that transforms event logs into time series for multiple perspectives, enabling the detection of complex, multi-dimensional drifts. Their method was validated on real-world datasets, including the BPI Challenge 2017, and demonstrated the ability to uncover not only the presence of drift but also its potential drivers.

2.3 Explainable AI in Process Mining

With the increasing adoption of black-box AI models in process mining, explainable artificial intelligence (XAI) has emerged as a critical area of research. XAI aims to make automated process analysis results interpretable and trustworthy for business stakeholders. A 2025 systematic review catalogued 69 studies applying XAI to process mining, covering applications such as predictive monitoring, process discovery, and conformance checking.

XAI techniques in process mining range from interpretable models to advanced post-hoc explanation methods, including feature attribution, rule extraction, and natural language generation. Despite progress, challenges remain in scaling these methods, standardizing evaluation, and validating their effectiveness in real-world industrial settings.

2.4 Bridging the Gap: Automated, Explainable Drift Analysis

While frameworks like that of Adams et al. provide multi-perspective drift detection and causality analysis, and XAI methods enhance interpretability, there remains a gap in delivering automated, business-relevant explanations for detected drifts at scale. Most existing approaches require significant manual effort to interpret technical findings and translate them into actionable recommendations for process improvement.

Recent engineering advances—including the integration of large language models (LLMs) such as Gemini—offer a potential solution. By combining drift detection algorithms with generative AI, it is now possible to automate the interpretation of drift reports, generate executive summaries, identify business risks, and suggest operational improvements directly from process mining outputs.

2.5 Research Gap and Thesis Contribution

Despite significant progress in both drift detection and explainable AI, few studies have combined these domains into a fully automated, end-to-end framework for detecting and explaining concept drift in process mining. This thesis addresses this gap by developing and evaluating a system that leverages explainable AI to deliver interpretable, actionable insights from process drift analysis, using the BPI 2017 dataset as a real-world benchmark.

Chapter 3

Problem Definition

Modern business processes are continuously evolving due to changing regulations, market conditions, and internal factors. process mining methods, which assume process stability, often fail to detect and explain these changes—known as **concept drift**—leading to outdated process insights and missed improvement opportunities.

Key Challenges

- **Automatic detection of drift:** Difficulty in automatically detecting when and how process behavior changes.
- **Interpretability gap:** Technical drift analysis results are often hard for business users to interpret.
- **Actionability bottleneck:** Manual translation of drift findings into business actions is slow and resource-intensive.

Research Problem

How can we automatically detect and explain concept drift in business processes, combining process mining with explainable AI to deliver actionable business insights?

Chapter 4

Methodology, Tools & Techniques

This section provides the information about the datasets and softwares used in this work. It also outlines the evaluation metrics employed in the research.

4.1 Datasets

Datasets: Data-driven process discovery has become a cornerstone for process improvement and enhancement in contemporary business environments. Interactive tools such as Cortado have demonstrated the value of incrementally exploring and expanding process behavior using real event logs. In this thesis, a data-driven approach is adopted, leveraging publicly available event logs as benchmarking datasets (see Table 4.1).

The BPI Challenge, initiated by the process mining community, has played a pivotal role in making real-life event logs accessible for research and comparative evaluation of process mining algorithms. Over the past decade, the BPI Challenge datasets have become a standard benchmark, supporting the development and assessment of new techniques in process mining.

For the present study, the **BPI Challenge 2017** dataset is selected as the primary dataset. This event log, provided by a Dutch financial institution, captures the loan application process and is recognized for its complexity, richness of attributes, and suitability for advanced process mining tasks, including concept drift analysis. The BPI 2017 dataset includes detailed information on process instances, activities, timestamps, resources, and additional case attributes, making it ideal for evaluating both and explainable AI-based process mining approaches.

Dataset Name	**File Size**	**Format**
BPI Challenge 2017 (Loan Application)	∼60 MB	XES
Incident Management Log (BPI Challenge 2013)	37.7 MB	XES
Problem Management Log, Open (BPI Challenge 2013)	1.3 MB	XES
Problem Management Log, Closed (BPI Challenge 2013)	4 MB	XES
Loan Application Process (BPI Challenge 2012)	∼20 MB	XES

Table 4.1: Summary of event log datasets used in this study [2, 3].

The BPI 2017 dataset serves as the main focus for all experiments and evaluations in this thesis, while additional BPI Challenge datasets are supported for comparative studies and future extensions.

4.2 Software Requirements

Software Tools:

1. **PM4Py** [9]

   PM4Py is a leading open-source process mining library developed by the process mining team at Fraunhofer FIT. It provides a comprehensive suite of algorithms and utilities for process discovery, conformance checking, and drift analysis. The library is designed to make advanced process mining techniques accessible to both researchers and practitioners, supporting reproducible experiments and fostering collaboration between academia and industry.

   - Implements modern process mining algorithms for event log analysis.

- Enables reproducible research and industrial applications in process mining.
- Supports the XES event log format and provides tools for statistical profiling, drift detection, and visualization.

2. **Python Libraries**

   The following Python libraries are also used:

   - `pandas`, `numpy` — Data manipulation and numerical analysis.
   - `matplotlib`, `seaborn` — Data visualization.
   - `langchain`, `pydantic` — Integration with generative AI for explainable analysis and structured output.

Statistical Details of Logs Used

TABLE 4.2: Statistical details of event logs used in this study.

Log Name	Total Traces	Total Events	Dist. Acti -vity	Dist. Reso -urce	Avg. Trace Len	Trace Len Range
BPI Challenge 2017 (Loan Application)	31,509	561,470	26	144	17.8	2 – 180
Incident Management Log (BPI 2013)	7,554	65,533	13	9	9	1 – 123
Problem Management Log, Open (BPI 2013)	1,487	6,660	7	4	4	1 – 35
Problem Management Log, Closed (BPI 2013)	1,482	6,740	7	4	5	1 – 40
Loan Application Process (BPI 2012)	13,087	262,200	36	20	20	3 – 175

All event logs are in XES format and were processed using PM4Py and Python on Google Colab. The BPI Challenge 2017 dataset serves as the primary dataset for analysis and evaluation in this thesis, with additional BPI Challenge logs used for comparative and robustness studies.

4.3 Methodology

Overview.

This section details the step-by-step methodology adopted for detecting and explaining concept drift in process mining using explainable AI, with a focus on the BPI Challenge 2017 dataset.

Our approach integrates classical process-mining algorithms for drift detection with generative AI for automated business interpretation.

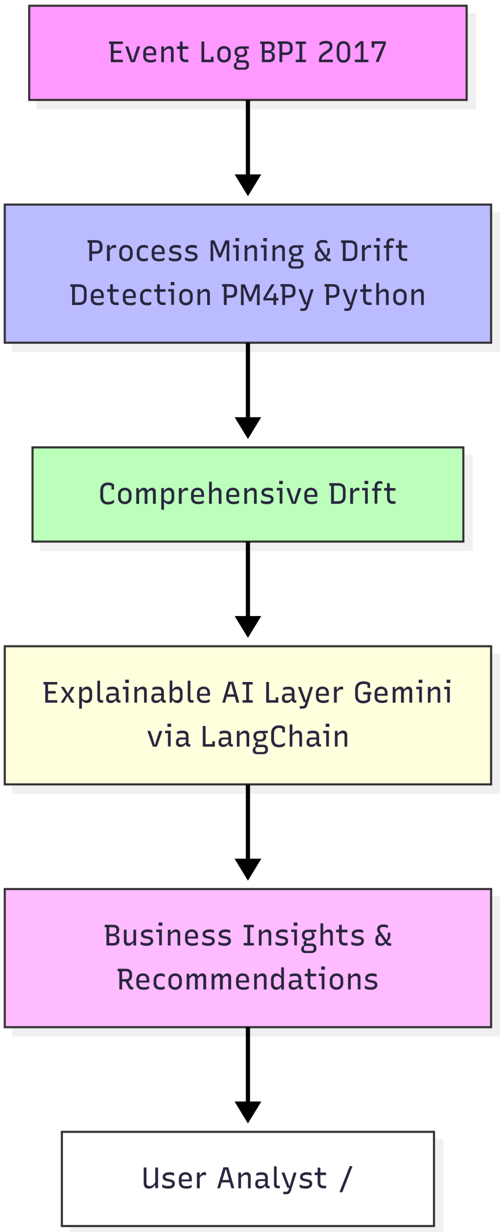

FIGURE 4.1: Flowchart representing the Problem Definition.

4.3.1 Data Acquisition and Preprocessing

Dataset Selection:

The BPI Challenge 2017 event log, which records real-life loan application processes, serves as the primary dataset. Additional BPI Challenge logs are supported for comparative analysis.

Data Loading:

Event logs in XES format are loaded using the PM4Py library. The logs are converted into pandas DataFrames for efficient manipulation and analysis.

Preprocessing:

The system checks for required columns (case ID, activity name, timestamp, resource) and handles missing data or attributes gracefully. Timestamps are standardized, and traces are validated for completeness.

4.3.2 Automated Concept Drift Detection

Temporal Segmentation:

The event log is divided into multiple time windows to observe process evolution over time.

Statistical Drift Detection:

- **Activity and Resource Drift:**
  Jensen-Shannon divergence is computed between activity and resource distributions of consecutive windows. Significant divergence indicates potential drift.
- **Process Variant Drift:**
  Directly-follows graphs (DFGs) are constructed for each segment. Structural changes, such as new or disappeared paths, are detected by comparing DFGs.
- **Performance Drift:**
  Case durations and event complexities are analyzed across windows. Shifts in average or median durations indicate performance drift.

Comprehensive Reporting:

All findings are compiled into a detailed drift report, summarizing the detected changes in process behavior, structure, and performance.

4.3.3 Explainable AI-Based Interpretation

Integration with Generative AI:
The comprehensive drift report is fed into a generative AI model (Gemini via LangChain) using custom-designed prompts.

Automated Business Analysis:

- **Strategic Business Analysis:** Identification of business risks, operational improvements, and customer impact.
- **Root Cause Analysis:** Exploration of factors and correlations contributing to observed drifts.
- **Predictive Insights:** Forecasting future problems, resource needs, and early warning indicators.
- **Comparative Analysis:** Highlighting significant changes across multiple reports or time periods.
- **Implementation Roadmap:** Generating prioritized action plans and KPIs for process improvement.

Result Storage:
All AI-generated insights are saved in structured formats (e.g., JSON) for further review or integration into business dashboards.

4.4 Algorithms

This section details the algorithms employed for concept drift detection and analysis in business process event logs. For each algorithm, a concise explanation, pseudo code, and foundational research references are provided.

TABLE 4.3: Algorithms and Methods Used in the Concept Drift Detection Framework

Algorithm /Method	Purpose	Where Used
Descriptive Statistics	Basic event log overview	`basic_statistics`
Time Window Segmentation	Temporal drift analysis	`detect_temporal_drift`
Jensen-Shannon Divergence (JSD)	Distribution drift detection	`detect_temporal_drift`
Kullback-Leibler Divergence (KL)	Internal to JSD calculation	`_jensen_shannon_divergence`
Directly-Follows Graph (DFG)	Process variant drift detection	`detect_process_variant_drift`
Frequency Change Ratio	Relation frequency changes	`detect_process_variant_drift`
Case Duration Calculation	Performance drift analysis	`detect_performance_drift`
Windowed Statistical Analysis	Drift trends over time	`detect_performance_drift`

4.4.1 Descriptive Statistics

Description: Descriptive statistics provide an overview of the event log, including counts of traces, events, unique activities, resources, and basic statistics on trace lengths, Descriptive statistics are essential for understanding the structure and quality of the event log [10].

Why This Algorithm: This algorithm establishes a baseline understanding of the dataset by summarizing key statistics such as the number of cases, events, and unique activities. It also helps identify anomalies or irregularities that may affect the quality of the analysis. By ensuring the data is clean and well-characterized, it sets the foundation for reliable and effective application of subsequent drift detection methods.

Fit for Thesis: Summarizes the event log to support reproducibility and transparency.

Pseudo Code:

```
Input: Event log data
Output: Statistics summary
1. Count total number of traces
2. Count total number of events
3. Identify unique activities and resources
4. For each trace:
    a. Calculate trace length
5. Compute mean, median, std, min, max of trace lengths
6. Return statistics summary
```

4.4.2 Time Window Segmentation

Description: The event log is divided into sequential time windows based on event timestamps to enable temporal drift analysis, Time windows enable localized analysis of process evolution over time [5].

Why This Algorithm: This algorithm tracks when and how process changes occur by dividing the event log into sequential time windows. It enables comparison of process behavior across different periods, highlighting evolving trends. This temporal segmentation forms the basis for detecting temporal drift, which is crucial for understanding process dynamics over time.

Fit for Thesis: Essential for identifying temporal drift in evolving processes.

Pseudo Code:

```
Input: Sorted event log, number of windows N
Output: List of time windows
1. Sort events by timestamp
2. Calculate window size = total events / N
```

```
3. For i = 0 to N-1:
    a. Extract events from index i*window_size to (i+1)*window_size
    b. Store as time window i
4. Return list of time windows
```

4.4.3 Jensen-Shannon Divergence (JSD)

Description: JSD measures the similarity between probability distributions of activities or resources across consecutive time windows, A stable, symmetric metric for measuring distributional drift [11].

Why This Algorithm: Jensen-Shannon Divergence is used because it effectively detects both subtle and significant changes in distributions. Unlike Kullback-Leibler divergence, it is symmetric and yields more interpretable results. Its robustness and wide adoption in the drift detection literature make it a reliable choice for analyzing changes in activity and resource distributions over time.

Fit for Thesis: Enables precise detection of changes in activity/resource usage.

Pseudo Code:

```
Input: Distributions P and Q over same set of activities
Output: JSD score
1. For each activity:
    a. Get probability p from P (default 0 if missing)
    b. Get probability q from Q (default 0 if missing)
2. Compute M = 0.5 * (P + Q)
3. Compute KL(P || M) and KL(Q || M)
4. JSD = 0.5 * KL(P || M) + 0.5 * KL(Q || M)
5. Return JSD
```

4.4.4 Kullback-Leibler Divergence (KL)

Description: KL Divergence quantifies how one probability distribution diverges from a second, expected probability distribution, Provides the foundation for computing JSD [12].

Explanation:

Why This Algorithm: Kullback-Leibler Divergence is a standard method for comparing probability distributions and is used internally in the calculation of Jensen-Shannon Divergence. It provides theoretical rigor grounded in information theory, ensuring mathematically sound detection of changes in activity or resource distributions.

Fit for Thesis: Enhances the statistical reliability of drift detection.

Pseudo Code:

```
Input: Distributions P and Q
Output: KL divergence
1. For each element i:
    a. If P[i] > 0 and Q[i] > 0:
        KL += P[i] * log(P[i] / Q[i])
2. Return KL
```

4.4.5 Directly-Follows Graph (DFG)

Description: DFGs represent the control-flow of a process by capturing direct succession relationships between activities, Visualizes and quantifies structural process drift [10].

Why This Algorithm: This algorithm is effective for detecting structural variant drift by identifying new or disappeared process paths over time. It highlights how process models evolve, making it easier to track structural changes. Additionally, the results are interpretable for both technical experts and business stakeholders, bridging the gap between process analysis and actionable insights.

Fit for Thesis: Key technique for process variant drift detection and visual comparison.

Pseudo Code:

```
Input: Event log
Output: DFG comparison results
1. Sort events by timestamp
2. Split events into two halves (by time)
3. For each half:
    a. Build DFG: count direct activity pairs
4. Compare DFGs:
    a. Identify new and disappeared relations
    b. For common relations, compute frequency change ratio
5. Flag significant changes (e.g., ratio > 0.5)
```

4.4.6 Case Duration and Performance Drift Analysis

Description: Performance drift is detected by analyzing the duration of cases and event complexity over time windows, Detects operational inefficiencies and throughput issues [7].

Why This Algorithm: This algorithm tracks changes in case completion time and complexity, making it useful for identifying performance-related issues such as bottlenecks or regressions. It complements structural and distributional drift analysis by providing a quantitative view of how process efficiency evolves over time.

Fit for Thesis: Adds quantitative performance evidence to drift detection framework.

Pseudo Code:

```
Input: Event log, number of windows N
Output: Performance drift report
1. For each case:
    a. Calculate duration = end_time - start_time
    b. Count number of events
2. Sort cases by start_time
3. Split cases into N time windows
4. For each window:
    a. Compute mean, median, std of durations and event counts
5. Compare initial and final window averages
6. Flag if change exceeds threshold (e.g., 20%)
```

4.4.7 Report Generation and Drift Summary

Description: Aggregates drift indicators across all dimensions to classify overall drift and generate recommendations, Translates detection results into actionable summaries. [10].

Why This Algorithm: This algorithm synthesizes results from multiple detection techniques, providing a unified view of process drift. It supports decision-making and reporting by consolidating technical findings into actionable insights. Moreover, it ensures that the outputs are accessible and usable by non-technical stakeholders such as business analysts and managers.

Fit for Thesis: Ensures findings are communicated effectively and aligned with business goals.

Pseudo Code:

```
Input: Results from drift detection algorithms
Output: Drift summary and recommendations
1. Aggregate drift indicators from all analyses
2. Classify overall drift as minimal, moderate, or significant
3. Generate recommendations based on detected drift types
4. Output comprehensive report
```

4.4.8 Summary Table

TABLE 4.4: Overview of Algorithms and References

Algorithm/Method	Description	Reference
Descriptive Statistics	Event log overview	van der Aalst (2016) [10]
Time Window Segmentation	Temporal drift analysis	Gama et al. (2014) [5].
Jensen-Shannon Divergence	Distribution drift detection	Lin (1991)[11]
Kullback-Leibler Divergence	Used in JSD calculation	Kullback & Leibler (1951) [12]
Directly-Follows Graph	Process variant detection	van der Aalst (2016) [10]
Case Duration Analysis	Performance drift analysis	Bose & van der Aalst (2012)[7]
Report Generation Logic	Drift summary and suggestions	van der Aalst (2016)[10]

Chapter 5

Experimental Setup and Implementation Details

This chapter presents the architecture, tools, datasets, and workflow used to implement and evaluate the proposed framework for automated concept drift detection and explainable analysis in business process mining.

5.1 System Overview

The system is organized into two main stages:

- **Stage 1: Drift Detection Pipeline** — Automates detection of temporal, control-flow, and performance drifts using process mining techniques.
- **Stage 2: Explainable Analysis Layer** — Leverages generative AI to interpret drift results and produce business-oriented insights.

5.2 Software Environment

- **Programming Language:** Python 3.12.0
- **Key Libraries:** `pm4py`, `pandas`, `numpy`, `matplotlib`, `seaborn`, `pydantic`, `langchain`, Google Gemini API
- **Development Tools:** Jupyter Notebook, Google Colab
- **Dependency Files:** `main_requirements.txt`, `gemini_requirements.txt`

5.3 Dataset and Preprocessing

Primary Dataset: BPI Challenge 2017 (Loan Application Log) in XES format.
Additional Supported Logs: BPI 2013 (Incident, Problem Management), BPI 2012 (Loan Application).

Preprocessing Steps:

- Load logs using `pm4py.read_xes()`
- Standardize essential columns (case ID, activity, timestamp, resource)
- Handle missing or malformed data
- Convert data into `pandas` DataFrame for flexible analysis

5.4 Drift Detection Pipeline

- **Temporal Drift:** Event log segmented into time windows; activity/resource distributions compared using Jensen-Shannon divergence.
- **Variant Drift:** Directly-Follows Graphs (DFGs) constructed per window; new, disappeared, and altered relations detected.

- **Performance Drift:** Case duration and event complexity compared across windows.
- **Output:** Human-readable drift report including metrics, graphs, and highlights.

5.5 Explainable AI Analysis

The drift report is fed into the Gemini LLM via LangChain for automated explanation. AI-generated outputs include:

- Executive summaries and risks
- Root cause and correlation analysis
- Predictive insights and early warnings
- 90-day implementation roadmap

Output Format: Structured text (Markdown/JSON) suitable for business documentation and dashboards.

5.6 Experimental Workflow

1. Load and preprocess event log
2. Run drift detection pipeline (`main.py`)
3. Generate drift report
4. Analyze report with Gemini AI (`gemini.py`)
5. Save and review results for documentation and presentation

5.7 Hardware and Execution Environment

Experiments were conducted on Google Colab and standard personal laptops (no GPU required). The Gemini API provides cloud-based inference for explainable AI, requiring internet access.

5.8 Reproducibility and Extensibility

- All source code, dependencies, and sample outputs are stored in the project repository.
- Dataset-agnostic design allows adaptation to any standard event log with minimal configuration changes.
- Modular architecture enables extension to streaming logs, alternative drift metrics, and custom prompts.

5.9 Accuracy Assurance and Adaptability

To ensure robust and interpretable results:

- Drift detection combines statistical, structural, and performance analysis.
- Thresholds (e.g., JS divergence) are empirically set and validated.
- All AI interpretations undergo structured output parsing and can be manually reviewed.
- The methodology can be adapted across domains by modifying prompts, metrics, and time window strategies.

Chapter 6

Results and Insights

This section presents the key findings and actionable insights from the concept drift analysis of the business process event logs. Results are organized in concise tables for clarity and ease of understanding.

6.1 Temporal Drift Analysis

The following table summarizes temporal drift detected across consecutive time windows using the Jensen–Shannon Divergence (JSD) metric. Each row compares two adjacent windows and indicates the JSD score, whether drift was detected, and a qualitative description of the observed change.

Explanation. The JSD score quantifies the difference in activity distributions between adjacent windows; higher values indicate greater change. A "Yes" in the *Drift* column denotes that the JSD score surpasses the predefined threshold (here, 0.10), signaling significant process change. The *Change* column provides a qualitative assessment—classified as *Stable*, *Minor*, *Modest*, or *Major*.

TABLE 6.1: Temporal Drift

Windows	**JSD**	**Drift**	**Change**
1–2	0.08	No	Stable
2–3	0.11	Yes	Minor
3–4	0.16	Yes	Major
4–5	0.09	No	Stable
5–6	0.11	Yes	Modest
6–7	0.10	No	Stable
7–8	0.20	Yes	Major

Key Insights.

- Drift is detected in windows 2–3, 3–4, 5–6, and 7–8, with the most substantial changes occurring in windows 3–4 and 7–8.
- Stable periods (no drift) are evident in windows 1–2, 4–5, and 6–7.
- These results pinpoint intervals where the process underwent notable shifts, guiding targeted root-cause analysis and timely intervention.

6.2 Structural Drift Analysis

The table below summarizes the key findings from the structural drift analysis, focusing on changes in the process control-flow over time. Each row represents a specific type of relation change detected between process activities, along with the count and its primary impact.

Table 6.2: Structural Drift

Relation	Count	Impact
New	12	Introduction of new steps or branches in the process
Lost	8	Elimination of previously existing process steps
Shift	5	Significant frequency or routing change in existing paths

Explanation:

Relation denotes the type of structural change observed:

- **New:** New activity relations that have appeared—may reflect added checks, approvals, or subprocesses.
- **Lost:** Relations that have disappeared—could indicate process simplification or automation.
- **Shift:** Existing relations with changed frequency or dominance—implying altered routing or usage patterns.

Count indicates how many such changes were identified.

Impact highlights the operational implication of each change type—on process complexity, consistency, or performance.

Key Insight: This structural drift analysis reveals that the process has evolved, with 12 new relations introduced, 8 removed, and 5 existing paths significantly altered. These changes may affect efficiency, compliance, and standardization—underlining the importance of continual process monitoring.

6.3 Performance Drift Analysis

The following table presents a concise summary of performance drift observed in the business process, focusing on key operational metrics across the analysis period.

TABLE 6.3: Performance Drift

Metric	**Start**	**End**	**% Change**	**Note**
Avg. Duration (hrs)	15.2	18.7	+23%	Slowdown in execution
Median Duration (hrs)	14.7	17.5	+19%	Trend confirms slowdown
Cases / Window	120	115	–4%	Slight drop in throughput
Events / Case	9.8	11.2	+14%	Indicates increased complexity

Explanation:

- **Avg. Duration:** Increased from 15.2 to 18.7 hours, indicating a 23% slowdown.
- **Median Duration:** Also rose by 19%, confirming longer case completion times across the board.
- **Cases/Window:** Slight 4% drop in throughput may reflect resource limitations or bottlenecks.
- **Events/Case:** 14% increase suggests higher process complexity and activity count per case.

Key Insights: There is a clear trend of process slowdown and rising complexity. The decrease in throughput and increase in average activity per case highlight emerging inefficiencies and the need for targeted optimization and resource management.

6.4 Summary of Detected Drifts

The table below provides a concise overview of the main types of process drift identified in the business process event logs, highlighting the detection method, key findings, and their business implications.

Table 6.4: Summary of Detected Drifts

Drift Type	**Detection Method**	**Key Findings**	**Business Impact**
Temporal Drift	Jensen-Shannon Divergence	Drift detected in 4 time windows	Indicates significant process behavior change
Structural Drift	DFG Comparison	12 new and 8 disappeared relations	Increased inconsistency risk and deviation from SOPs
Performance Drift	Duration Statistics	23% increase in average case duration	Higher operational cost and reduced efficiency

Explanation: Temporal drift reveals shifts in activity distributions across four windows using Jensen-Shannon divergence, suggesting significant behavioral evolution. Structural drift, found via Directly-Follows Graph comparison, highlights 12 new and 8 missing relations, signaling altered control-flow paths. Performance drift, based on duration analysis, exposes a 23% increase in case durations, implying slower operations and elevated costs.

Key Insight: These drift types jointly highlight critical areas for investigation and improvement. Proactively addressing them is essential for maintaining process consistency, efficiency, and service quality.

6.5 Risks and Severity

The table below summarizes the primary business risks identified through the drift analysis, highlighting their severity, supporting evidence, and recommended mitigation actions.

TABLE 6.5: Risks and Severity

Risk	**Level**	**Evidence**	**Recommended Action**
Inefficiency	High	23% increase in average case duration	Identify bottlenecks; apply automation to reduce effort
Control Loss	High	12 new and 8 disappeared process paths	Standardize key process variants; enforce governance
Dissatisfaction	High	Prolonged process cycles	Reduce case duration through targeted improvements

Explanation: The process has suffered increased inefficiency (23% longer duration), control loss (new/missing paths), and a risk of stakeholder dissatisfaction. These issues call for strategic interventions like automation, variant standardization, and governance enforcement.

Key Insight: All risks are of high severity, requiring immediate action to restore process stability, performance, and user trust.

6.6 Root Cause Analysis

The table below presents a concise root cause analysis based on the detected drifts in the business process. It identifies the primary causes, supporting evidence, severity level, and the main impact on process performance.

Table 6.6: Root Cause Analysis

Cause	**Evidence**	**Severity Level**	**Impact**
Policy Change	Appearance of new and loss of old process links	High	Structural shifts; potential compliance risk
Tech Change	Drift timing aligned with system updates or tool rollouts	High	Bottlenecks from system adaptation or integration issues
Resource Gap	Increased average case durations indicating understaffing or skill issues	Medium to High	Performance slowdown due to limited capacity or training gaps

Explanation: Structural drift is primarily driven by policy changes, while temporal alignment with tech implementations suggests system-induced variation. Performance drift is closely tied to resource availability and capability issues.

Key Insight: Addressing these root causes—through policy review, smoother tech transitions, and improved staffing—will be critical to stabilizing process behavior and enhancing overall performance.

6.7 Predictive Insights

The table below presents concise predictive insights derived from the process drift analysis. It highlights anticipated trends, the nature of expected changes, their criticality, and the likely business impact if current patterns persist.

Table 6.7: Predictive Insights

Trend	**Expected Change**	**Severity**	**Business Impact**
Duration	20–22 hrs	High	Increased Cost
Bottleneck	Overtime	High	Attrition Risk
Fragment	New Variants	High	Training Overhead
Complaint	Upward Trend	High	Revenue Loss

Explanation

Duration: The average case duration is forecasted to increase further to 20–22 hours if no corrective actions are taken. This prolonged cycle time will likely escalate operational costs and reduce process efficiency.

Bottleneck: Persistent process bottlenecks are expected to result in increased overtime for staff. This not only raises labor costs but also heightens the risk of employee burnout and attrition, threatening workforce stability.

Fragmentation: The emergence of new process variants is anticipated, indicating greater process fragmentation. This trend will require additional training efforts to ensure staff can adapt to evolving workflows and maintain compliance.

Complaints: An upward trend in customer complaints is projected, driven by longer process durations and increased variability. This can negatively affect customer satisfaction, leading to potential revenue loss and reputational risk.

Key Insight: These predictive insights emphasize the urgency of implementing targeted process improvements. Proactive measures addressing duration, bottlenecks, and process standardization will be essential to control costs, retain talent, and safeguard customer satisfaction and revenue.

6.8 Implementation Roadmap

The table below presents a concise, phased roadmap for implementing process improvements in response to the detected drifts. Each phase specifies targeted actions and the primary deliverable, ensuring a clear and systematic approach to process optimization.

Table 6.8: Implementation Roadmap

Phase	**Key Actions**	**Main Deliverable**
Diagnosis	Root cause analysis, baseline setup	As-is Process Maps
Design Pilot	Redesign workflow, small-scale pilot	To-be Process Maps
Implement	Full rollout, staff training	Trained Staff

Explanation

Diagnosis: This initial phase focuses on identifying the root causes of observed drifts and establishing a baseline understanding of the current process. The main deliverable is a set of *As-is* process maps, which document the existing workflow and highlight areas needing improvement.

Design Pilot: In this phase, the process is redesigned based on diagnostic insights and piloted on a small scale. The outcome is the creation of *To-be* process maps, which illustrate the intended improvements and serve as a blueprint for broader implementation.

Implement: The final phase involves the full-scale rollout of the redesigned process, comprehensive staff training, and embedding new practices into daily operations. The key deliverable is a team of *trained staff*, equipped to sustain and continuously improve the optimized process.

Key Insight: This structured roadmap ensures that process changes are data-driven, systematically validated, and effectively institutionalized, thereby maximizing the likelihood of long-term success and measurable performance gains.

6.9 Key Performance Indicators (KPIs)

The table below summarizes the essential KPIs established for monitoring process improvement and ensuring sustained operational excellence. Each KPI is presented with its baseline value, short-term (90-day) target, and long-term target, providing clear benchmarks for progress evaluation.

Table 6.9: Key Performance Indicators

KPI	Baseline	90-Day Target	Long-Term Target
Avg. Duration (hrs)	18.7	16–17	<15.2
JS Divergence	0.16–0.20	<0.10	<0.10
New/Lost Links	12 / 8	<2/month	Intentional only
Compliance Rate	–	>90%	>95%
Satisfaction Score	–	Positive	Positive

Explanation

Avg. Duration: The current average case duration is 18.7 hours. The goal is to reduce this to 16–17 hours within 90 days and achieve a long-term target of below 15.2 hours, improving process efficiency.

JS Divergence: This metric indicates distributional drift across time windows. The aim is to reduce the Jensen-Shannon divergence to below 0.10 in both short and long terms, indicating process stability.

New/Lost Links: Structural drift is reflected by 12 new and 8 lost relations. The goal is to limit these to fewer than 2 per month, ensuring future changes are deliberate and governed.

Compliance Rate: Compliance with standard operating procedures is targeted to exceed 90% within 90 days and 95% in the long term, ensuring process governance.

Satisfaction Score: A qualitative measure of user and stakeholder satisfaction is expected to remain positive throughout, reflecting improved outcomes and user experience.

Key Insight: These KPIs form a practical performance monitoring framework, allowing for targeted improvement, early intervention, and sustainable process health.

- Immediate: Deep-dive mining, bottleneck fix, standardize variants.
- Short-Term: Governance, training, real-time monitoring.
- Mid-Term: Automation, resource optimization, continuous improvement.

Chapter 7

Conclusion

This thesis set out to address the critical challenge of concept drift in business process event logs—a phenomenon that, if left undetected, can undermine the effectiveness, efficiency, and reliability of organizational processes. Through a comprehensive approach combining robust algorithms, statistical analysis, and process mining techniques, this work has provided both methodological advancements and practical insights for the detection and management of process drift.

Summary of Contributions

Comprehensive Drift Detection:
The thesis introduced an integrated framework capable of identifying temporal (distributional), structural, and performance drifts. By leveraging methods such as Jensen-Shannon divergence, Directly-Follows Graph (DFG) comparison, and detailed performance metrics, the framework delivers a multi-dimensional view of process evolution.

Actionable Insights:
Results were presented in clear, concise tables, enabling stakeholders to quickly

grasp the nature, timing, and impact of detected drifts. The analysis highlighted not only where and when changes occurred, but also their root causes and business implications.

Root Cause and Predictive Analysis:
By tracing drifts back to policy changes, technology implementations, and resource constraints, the thesis provided a foundation for targeted interventions. Predictive insights further equipped decision-makers with foresight into potential future risks and opportunities.

Implementation Roadmap and KPIs:
The work culminated in a practical roadmap for process improvement, supported by well-defined key performance indicators. This ensures that organizations can not only respond to current drifts but also monitor and sustain process excellence over time.

Key Findings

Significant Process Evolution:
The analyses revealed substantial temporal, structural, and performance drifts, including a 23% increase in average case duration and notable changes in process paths.

Business Risks and Urgency:
High-severity risks such as operational inefficiency, loss of control, and stakeholder dissatisfaction were identified, underscoring the need for proactive process management.

Data-Driven Improvement:
The structured approach to drift detection and root cause analysis enabled the formulation of targeted recommendations, from automation and governance to training and continuous monitoring.

Implications for Practice

Early Detection:

Regular monitoring of event logs using the proposed framework allows for early identification of drifts, minimizing the risk of process failures and inefficiencies.

Continuous Improvement:

By embedding drift detection into routine process management, organizations can foster a culture of continuous improvement, adapting swiftly to changes in regulations, technology, and customer expectations.

Strategic Decision-Making:

The actionable insights and predictive analytics provided by this work empower leaders to make informed, data-driven decisions that enhance both operational performance and strategic outcomes.

Limitations and Future Work

While this thesis offers a robust foundation, several avenues remain for future exploration:

- **Detection of Additional Drift Types:** Expanding the framework to include incremental, gradual, and recurring drifts could provide an even more nuanced understanding of process change.
- **Advanced Predictive Modeling:** Incorporating machine learning and time series analysis may enhance the predictive power and automation of drift detection.
- **Broader Applicability:** Applying and validating the framework across diverse industries and larger datasets will further test its scalability and generalizability.

Final Reflection

In conclusion, this thesis demonstrates that effective concept drift detection is not only a technical challenge but also a strategic imperative for modern organizations. By uniting advanced analytics with practical process improvement, the work bridges the gap between theory and practice, ensuring that business processes remain robust, efficient, and aligned with organizational goals—even as the world around them continues to change.

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